Phase 2A: sharper Sidon density for perfect difference sets

An exploration

May 11, 2026

1 The question

For an infinite perfect difference set (PDS) $A \subset \mathbb{N}$, the classical Lindström bound gives

$$k(x) := |A \cap [1, x]| \;\leq\; \sqrt{x} + \sqrt[4]{x} + 1. \tag{1}$$

Singer-type constructions of Sidon sets achieve $k(x) = (1 + o(1))\sqrt{x}$ at sparse x. Cilleruelo–Nathanson [**?**] construct a *PDS* satisfying

$$\limsup_{x\to\infty} \frac{k(x)}{\sqrt{x}} \;\geq\; \frac{1}{\sqrt{2}}. \tag{2}$$

So the "Sidon ceiling" for PDS is at most 1 and at least $1/\sqrt{2}$.

Phase 2A target. Prove $\limsup k(x)/\sqrt{x} \leq 1 - \delta$ for some explicit $\delta > 0$ and all infinite PDS A — ideally with $1 - \delta = 1/\sqrt{2}$, matching the CiNa08 lower bound.

2 What is automatic

2.1 $\liminf k(x)/\sqrt{x} = 0$

The infinitely-often lower bound on a_n from Phase 2C gives a strong constraint on $\liminf k(x)/\sqrt{x}$.

Proposition 1. *For any infinite PDS A and every $c < 2$,*

$$\liminf_{x\to\infty} \frac{k(x)}{x^{1/(2c)}} \;\leq\; \sqrt{2}.$$

In particular $\liminf k(x)/\sqrt{x} = 0$*, and more sharply* $\liminf k(x) \cdot x^{-(1/4+\varepsilon)} = 0$ *for every* $\varepsilon > 0$.

Proof. By Theorem 2 of the companion note (the GPT-5.4 Pro partition argument), for every $c < 2$ there are infinitely many n with $a_n \geq n^c$. At such n, by the PDS counting identity (Lemma 1 of the companion note), $N(a_n) = \binom{k(a_n)}{2}$. Since $n \leq N(a_n)$ trivially, $\binom{k(a_n)}{2} \geq n$, so $k(a_n) \geq \sqrt{2n}$. But we also need the *upper* side: at the same a_n, $n = N(a_n) - O(1)$, so $k(a_n) \leq (1+\sqrt{1+8n})/2 \leq \sqrt{2n}+1$.

Combining with $a_n \geq n^c$, i.e. $n \leq a_n^{1/c}$, gives $k(a_n) \leq \sqrt{2a_n^{1/c}} + 1 = \sqrt{2}\, a_n^{1/(2c)} + 1$. So $k(a_n)/a_n^{1/(2c)} \leq \sqrt{2} + o(1)$. ☐

In words: at infinitely many $x = a_n$, the PDS density is bounded by $\sqrt{2} \cdot x^{1/(2c)}$ for any $c < 2$ — substantially below the Sidon ceiling $\sqrt{x}$. As $c \to 2$, the bound at those x approaches $x^{1/4}$.

2.2 What $\liminf = 0$ does *not* give

Proposition **??** controls the density at sparse x (the "hard" positions where a_n peaks), but says nothing about $\limsup k(x)/\sqrt{x}$. A PDS could in principle have $k(x)$ touching $\sqrt{x}$ at frequent intermediate x values while dropping low at the i.o. sparse positions. The Phase 2A target is the missing piece.

3 Attempt 1: Cauchy–Schwarz via the PDS partition identity

3.1 Setup

Let $A_M = A \cap [1, M]$ and $k = |A_M|$. Let $D_M := A_M - A_M$ (positive part), so $|D_M| = \binom{k}{2}$. Let T_M be the largest T with $[1, T] \subset D_M$ (the "covered prefix").

The PDS counting identity gives $N(M) = \binom{k}{2}$ where $N(M) = \#\{n : a_n \le M\}$. The covered prefix is $T_M = \#\{n \le N(M) : a_n \le M, b_n \le M\} = \#\{n : \text{both endpoints in } A_M\} = N(M)$. *Wait*: that's only correct if $a_n \le M \Rightarrow b_n \le M$, which is automatic since $b_n < a_n \le M$.

So actually $T_M = N(M) = \binom{k}{2}$: every $n \le N(M)$ has its representation in A_M. So $T_M = \binom{k}{2}$.

Hmm: but the differences of A_M include $\binom{k}{2}$ values all $\le M - 1$. The covered prefix $[1, T_M]$ has T_M distinct values, and they form a contiguous interval. The remaining $\binom{k}{2} - T_M$ differences are scattered in $(T_M, M - 1]$.

So $T_M \le \binom{k}{2} \le M - 1$, with $T_M = \binom{k}{2}$ in the "perfectly covered" case where every difference in $[1, \binom{k}{2}]$ is realised.

3.2 The constraint

For a PDS, every $n \in \mathbb{N}$ is realised exactly once. So at the truncation A_M, the realised differences are exactly $\{n : a_n \le M\} = [1, N(M)]$. Hence $T_M = N(M) = \binom{k}{2}$ *always.*

This is a stronger statement than "$T_M \le \binom{k}{2}$" — it says equality. So all $\binom{k}{2}$ differences of A_M lie in $[1, \binom{k}{2}]$, i.e. none in $(\binom{k}{2}, M - 1]$.

Now: the differences of A_M are at most $M - 1$. They're all in $[1, \binom{k}{2}]$. So we have $\binom{k}{2}$ distinct integers in $[1, \binom{k}{2}]$ — which forces them to be *exactly* $\{1, 2, \dots, \binom{k}{2}\}$.

Lemma 1. *Let A be an infinite PDS, M any positive integer, $k = k(M)$. Then* $A_M - A_M = \{-\binom{k}{2}, \dots, -1, 0, 1, \dots, \binom{k}{2}\}$ exactly.

Proof. Every $n \in \mathbb{Z}_{>0}$ has unique representation in A. The representation lies in A_M iff $a_n \le M$. The set of such n is $\{n : a_n \le M\}$, which equals $\{1, 2, \dots, N(M)\}$ since a_n is monotone increasing in n in the sense that $\{n : a_n \le M\}$ is a downward-closed set of indices.

Wait — is a_n monotone? NOT necessarily. The greedy data shows a_n is bimodal and far from monotone. So $\{n : a_n \le M\}$ need not be an initial segment. $\square$

Correction. The argument in Lemma **??** fails: a_n is not monotone, so $\{n : a_n \le M\}$ is not an initial segment $[1, N(M)]$. The set T_M I claimed equals $N(M) = \binom{k}{2}$ is actually the largest *contiguous prefix* of covered n, which is not the same as $|\{n : a_n \le M\}|$.

The correct relations:

- $|\{n : a_n \le M\}| = N(M) = \binom{k(M)}{2}$ (PDS counting).
- T_M = largest T with $[1, T] \subset A_M - A_M$. Generally $T_M \le N(M)$ with equality only if $\{n : a_n \le M\}$ is the initial segment $[1, N(M)]$ — which fails in general.

So my "perfect cover" observation is wrong. The correct statement is:

Lemma 2. $T_M \le N(M) = \binom{k(M)}{2}$.

This is just the trivial inequality, without yielding the structural conclusion I wanted.

3.3 What went wrong

The hope was that PDS forces A_M to be "maximally packed" — every short difference within $[1, M-1]$ realised. This fails because PDS only requires that every n *eventually* be realised, not by elements in A_M. So T_M can be considerably less than $\binom{k}{2}$.

In fact, for greedy at $N = 1898$, $k = 2131$, $\binom{k}{2} \approx 2.27 \times 10^6$, but $T_M = 1898$ (the greedy covers $[1, N]$ at scale $M = a_N$, but the $1898 - N(M)$ "extra" differences are scattered in $(N, M-1]$).

So the partition argument from Phase 2C is the right tool — attempting to use $T_M = \binom{k}{2}$ directly fails for non-greedy PDS.

4 Attempt 2: Fejér weighted Plancherel

4.1 The identity

For Sidon A_M, the Fejér-kernel inner product

$$I_T \;:=\; \int_0^1 K_T(\xi)\,|\widehat{1_{A_M}}(\xi)|^2\,d\xi$$

satisfies, expanding the autocorrelation,

$$I_T \;=\; r(0) + 2\sum_{n=1}^{T-1} r(n)\,(1 - n/T) \;=\; k + 2\sum_{n=1}^{T-1} r(n)(1-n/T),$$

where $r(n) = (1_{A_M} \star 1_{A_M}^{-})(n) \in \{0,1\}$ for $n \neq 0$ (Sidon) and equals 1 for $n \in [1, T_M]$ (PDS coverage).

For $T \le T_M$: $r(n) = 1$ for all $n \in [1, T-1]$, so

$$I_T \;=\; k + 2\sum_{n=1}^{T-1}(1 - n/T) \;=\; k + (T-1).$$

4.2 Upper bound on I_T

We want a k-dependent upper bound on I_T to pin down k.

Trivial bound. K_T has L^1 norm 1 (it integrates to 1), and $|\widehat{1_{A_M}}|^2 \le k^2$ pointwise. So $I_T \le k^2 \cdot 1 = k^2$. Together with $I_T = k + T - 1$: $T - 1 \le k^2 - k$, i.e., $T \le k^2 - k + 1 \approx \binom{k}{2} \cdot 2$. Same as Sidon, no improvement.

Tighter bound using $|\widehat{1}|^2$'s peak structure. The function $|\widehat{1_{A_M}}|^2$ peaks at $\xi = 0$ with value k^2, and has L^2 norm $\sqrt{2k^2 - k} \approx k\sqrt{2}$ (from $\int |\widehat{1}|^4 = 2k^2 - k$). It's also nonnegative.

If $|\widehat{1_{A_M}}|^2$ were close to $k^2 \cdot \nVdash_{\xi \approx 0}$ of width $1/k$, then $I_T \approx K_T(0) \cdot k^2 \cdot (1/k) = T \cdot k$, giving $T - 1 \approx Tk - k$, i.e. $T \approx 1 + Tk - k$, no useful bound.

If instead $|\widehat{1}|^2$ has heavy tail behavior, the bound is different. We do not see a clean way to exploit PDS beyond Sidon at this level.

5 Attempt 3: higher moments of the autocorrelation

5.1 L^p approach

For $p \geq 4$:

$$\int |\widehat{1}_{A_M}|^{2p} d\xi \;=\; \sum_n |(1_{A_M}^{\star p}) \star (1_{A_M}^{-\star p})(n)|.$$

For Sidon, this counts solutions to $a_1 - b_1 + a_2 - b_2 + \cdots = n$ with $a_i, b_i \in A_M$. The total ($n = 0$ case) gives

$$\int |\widehat{1}|^{2p} \;\geq\; |A_M|^p \cdot p! \,/\text{symmetry}.$$

Standard energy-style estimates apply.

For PDS, are these higher moments different from Sidon? *In general no*, because the PDS structure constrains *which* differences appear but not the total count or the higher-order solution count. Higher moments don't distinguish.

This is a discouraging result: at every moment level we've tried, PDS is no different from Sidon.

6 Sub-targets

The full Phase 2A target seems out of reach with elementary identities. Tractable sub-targets:

6.1 Sub-target A: limsup density at perfect scales

Conjecture 1. *For any infinite PDS A, the set of x with $k(x) > (1-\delta)\sqrt{x}$ has density zero in $\mathbb{N}$, for some $\delta > 0$.*

This is weaker than "$\limsup \leq 1-\delta$" but would still strengthen the picture: PDS can be dense, but only at isolated scales.

6.2 Sub-target B: gap between $\limsup$ and Sidon ceiling

CiNa08's $\geq 1/\sqrt{2}$ is the best lower bound on $\limsup k(x)/\sqrt{x}$ for PDS. Lindström's ≤ 1 is the best upper bound for Sidon (hence PDS). Closing this 0.293 gap is the natural problem.

Realistic attack: show that PDS at density $\geq (1/\sqrt{2}+\delta)\sqrt{x}$ infinitely often, combined with the i.o. lower bound $a_n \gg n^{2-o(1)}$, gives a contradiction. The intuition is that $\liminf k(x)/\sqrt{x} = 0$ very fast (Proposition **??**) should constrain how often k can be near the Sidon ceiling.

A first computation: if $k(x_j) \geq (1-\delta)\sqrt{x_j}$ at x_j, then $N(x_j) \geq (1-\delta)^2 x_j/2$. Combined with the i.o. lower bound $a_n \geq cn^c$ at $n_j = N(x_j)$:

$$x_j \geq a_{n_j} \geq c\, n_j^c \geq c\,((1-\delta)^2 x_j/2)^c.$$

For $c < 2$: $x_j \geq c\,(1-\delta)^{2c}\, 2^{-c}\, x_j^c$, so $x_j^{c-1} \leq 2^c/(c(1-\delta)^{2c})$, which fails for x_j large since $c > 1$. So the assumption "$k(x_j) \geq (1-\delta)\sqrt{x_j}$ infinitely often, combined with the i.o. lower bound at the SAME x_j" is impossible.

But this is exactly Proposition **??** rephrased: the i.o. lower bound automatically excludes density near $\sqrt{x}$ at the SAME x values where the lower bound triggers.

To get a uniform Phase 2A bound, we'd need to show that the i.o. lower bound triggers densely enough to dilute $\limsup k(x)/\sqrt{x}$. But the i.o. statement gives no upper bound on the gap between trigger points.

6.3 Sub-target C: density of a_n peaks

Sharpen the Phase 2C i.o. statement to: for every $c < 2$, the density of $\{n : a_n \geq c\, n^c\}$ in $\mathbb{N}$ is at least $\rho_c > 0$.

This would give: at *positive density* of n, a_n is large. Combined with density estimates, this could constrain $\limsup k(x)/\sqrt{x}$.

The Phase 2C argument as written gives the i.o. statement but *not* a positive-density statement. Strengthening it is its own open subproblem.

7 Conclusions

Phase 2A as initially framed — "$\limsup k(x)/\sqrt{x} \leq 1 - \delta$" — is genuinely hard with our current tools. We documented three attempts that all fail:

- Cauchy–Schwarz via the partition identity reduces back to Sidon.
- Fejér-kernel weighted Plancherel doesn't exploit PDS beyond Sidon.
- Higher L^p moments of $|\widehat{1_A}|$ are determined by Sidon alone.

The structural reason is consistent: the PDS-vs-Sidon distinction shows up in *which differences appear at small scales*, not in aggregate counting quantities. The Phase 2C partition argument is the unique tool we have for converting "which differences appear" into a quantitative bound, and it caps out at the $c = 2$ ceiling.

Three reasonable sub-targets remain:

1. Conjecture **??**: density-zero exceptional set.
2. Sub-target B: close the 0.293 gap between $1/\sqrt{2}$ and 1.
3. Sub-target C: positive-density version of Phase 2C.

We do not know how to attack any of (1)–(3) with elementary tools. Phase 2A is best regarded as identifying these sub-targets and *not* as a deliverable theorem.

References

[1] J. Cilleruelo and M. B. Nathanson, *Perfect difference sets constructed from Sidon sets*, Combinatorica 28 (2008), 401–414.